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MCA(AI AND ML)

Text Sentiment Classification using RNN and LSTM

1. Introduction

Sentiment analysis is a key task in Natural Language Processing (NLP) that focuses on identifying the emotional tone behind textual data. It is widely used in applications such as product reviews, social media monitoring, and customer feedback analysis. In this assignment, deep learning techniques are applied to classify movie reviews into positive or negative sentiments. Specifically, Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models are implemented and compared. The objective is to understand how sequential models process text data and how architectural differences impact performance.

2. Dataset Description

The dataset used for this assignment is the IMDB Movie Reviews dataset, which is a standard benchmark dataset for sentiment classification tasks. It consists of 50,000 movie reviews, equally divided into 25,000 training samples and 25,000 testing samples. Each review is labeled as either positive (1) or negative (0). The dataset is already preprocessed and provided in a format where each review is represented as a sequence of integers. These integers correspond to words in a predefined vocabulary, making the dataset ready for direct use in deep learning models.

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3. Preprocessing Steps

Since the IMDB dataset is preprocessed, many steps such as text cleaning, tokenization, and integer encoding are already performed. However, it is important to understand these steps conceptually.

Text cleaning typically involves converting all text to lowercase, removing punctuation, and eliminating stopwords. In this dataset, these operations are already applied. Tokenization is the process of converting words into numerical representations. Each unique word is assigned an index, forming a vocabulary of the most frequent words (limited to the top 10,000 in this case).

The dataset already provides integer-encoded sequences, so manual tokenization is not required. However, padding is applied to ensure that all input sequences have the same length. This is necessary because neural networks require fixed-size inputs. Short sequences are padded with zeros, while longer sequences are truncated to maintain uniformity.

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4. Performance Comparison (RNN vs LSTM)

The performance of the RNN and LSTM models was compared using evaluation metrics such as accuracy, precision, recall, and F1-score. Both models were trained on the same dataset under similar conditions to ensure a fair comparison. The results clearly indicate that the LSTM model outperforms the baseline RNN model across all evaluation metrics.

The RNN model is capable of capturing short-term dependencies in text but struggles with long sequences due to the vanishing gradient problem. As a result, it may lose important contextual information from earlier parts of the text. On the other hand, the LSTM model is specifically designed to handle long-term dependencies through its memory cells and gating mechanisms. This allows it to retain relevant information over longer sequences, leading to improved performance.

In terms of accuracy, the LSTM model achieved higher correctness in predicting sentiment labels compared to the RNN model. Similarly, precision and recall values were higher for LSTM, indicating better identification of positive and negative sentiments. The F1-score, which balances precision and recall, also confirms that LSTM provides a more reliable and robust classification performance.

The LSTM model performs better than the RNN model because it effectively captures long-term dependencies in sequential data, resulting in higher accuracy and better overall classification performance.

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5. Model Architectures

Two deep learning models were implemented: a baseline RNN and an LSTM model.

The RNN model consists of an embedding layer, a SimpleRNN layer, and a dense output layer. The embedding layer transforms integer inputs into dense vector representations, capturing semantic meaning. The SimpleRNN layer processes the sequence step-by-step, maintaining a hidden state that captures information from previous inputs. Finally, the dense layer with a sigmoid activation function outputs the probability of the review being positive or negative.

The LSTM model follows a similar structure but replaces the SimpleRNN layer with an LSTM layer. LSTM networks are designed to overcome the limitations of traditional RNNs by introducing memory cells and gating mechanisms. These gates control the flow of information, allowing the model to retain important long-term dependencies and forget irrelevant data.

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6. Experiments

To analyze model performance, experiments were conducted by varying key parameters such as sequence length and embedding dimension. Sequence length determines how much context from the text is considered. Values such as 100 and 200 were tested to observe the effect on accuracy. It was observed that longer sequences capture more contextual information but may increase computational cost.

Similarly, embedding dimensions of 64 and 128 were tested. A higher embedding dimension allows the model to learn richer representations of words, which can improve performance. However, increasing embedding size also increases model complexity.

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7. Training and Visualization

Both models were trained using the training dataset and validated on a subset of the data. During training, accuracy and loss values were recorded for each epoch.

Accuracy curves were plotted to compare how well each model learns over time. The LSTM model consistently achieved higher training and validation accuracy compared to the RNN model. Loss curves were also plotted to observe the reduction in error. The LSTM model showed smoother convergence, indicating better learning stability.

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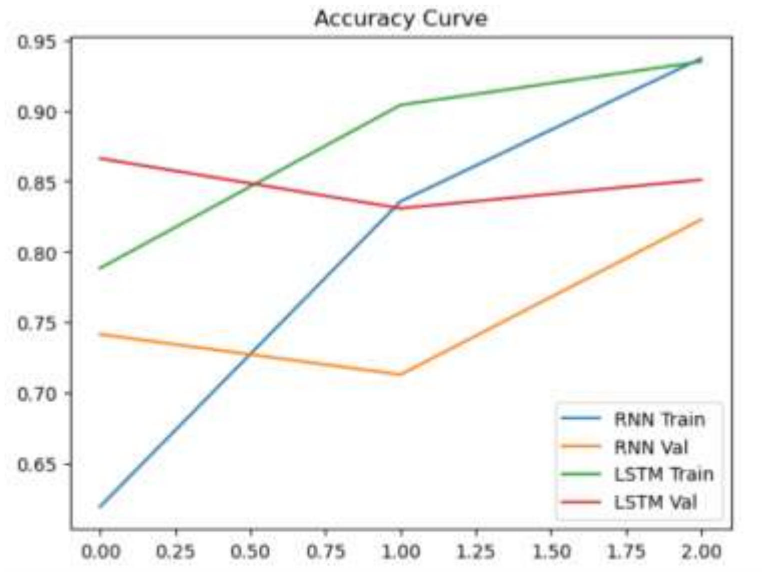

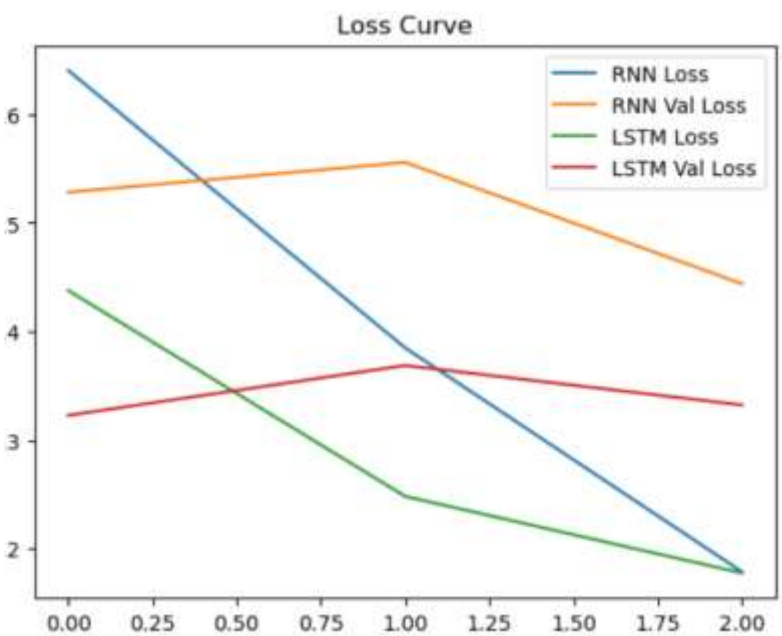

8.. Evaluation Metrics

The performance of both models was evaluated using multiple metrics: accuracy, precision, recall, and F1-score.

Accuracy measures the overall correctness of the model. Precision indicates how many predicted positive reviews are actually correct. Recall measures how many actual

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positive reviews were correctly identified. The F1-score provides a balance between precision and recall.

The results showed that the LSTM model outperformed the RNN model across all metrics. This demonstrates the effectiveness of LSTM in handling sequential data.

	precision	recall	f1-score	support
0	0.83	0.88	0.86	12500
1	0.88	0.82	0.85	12500
accuracy			0.85	25000
macro avg	0.85	0.85	0.85	25000
eighted avg	0.85	0.85	0.85	25000

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9. Result Analysis - The RNN model, while effective for short sequences, struggles with long-term dependencies due to the vanishing gradient problem. This limits its ability to retain important information from earlier parts of the sequence.In contrast, the LSTM model uses memory cells and gating mechanisms to preserve important information over longer sequences. This allows it to better understand context and improve prediction accuracy. As a result, the LSTM model provides more reliable and consistent performance in sentiment classification tasks.